By TOUCH DEN · November 11, 2026

The CIA Triad and Beyond

Video 1 · The Three Core Goals

Security Fundamentals · Section 2

In this video, you will learn:

- Why security needs clear, named goals
- Confidentiality, integrity, and availability
- A quick test for each one

Why we need clear goals

"Make it secure" is not a plan. Secure against what? Readable by whom? Working how often?

- *Without named goals, two people both say "secure" and mean different things, and neither can check it.*
- *Security people use a small set of named goals, so you can **design for**, **test**, and **detect the failure of** each one.*
- *The oldest and most useful set is three: the **CIA triad**.*

(The name is just the first letters. It has nothing to do with any agency.)

Confidentiality

Only the right people can _read_ the information.

- *Broken by a **data breach**: a stolen laptop, an open database, an email to the wrong person, traffic read on open Wi-Fi.*
- *Main tools: **encryption**, **access control**, **authentication**.*

***Quick test:** if this fell into a stranger's hands, would it matter? If yes, it needs protecting.*

Integrity

The information is correct, and _changed_ only by the right people, in the right way.

- *Covers **malicious** change (an attacker alters a payment) and **accidental** change (a bug, a faulty disk).*
- *Main tools: **hashing** (a fingerprint that changes if the data changes), **digital signatures**, **validation**.*

***Quick test:** if someone changed this without permission, who would be harmed, and would we even notice?*

Availability

The system _works_ when the people who need it need it.

- *Secret and correct data is useless if you cannot reach it (a hospital record in an emergency).*
- *Broken by ransomware, a flood of fake traffic, a power cut, a failed disk. Often harmed by accident, not only attackers.*
- *Main tools: **backups**, **redundancy**, **capacity and filtering**.*

***Quick test:** if this were unreachable for an hour, a day, a week, what would it cost?*

Recap: the three core goals

- *Named goals let us design, test, and detect failures.*
- ***Confidentiality:** who can read it. **Integrity:** who can change it. **Availability:** does it work when needed.*
- *Integrity covers accidents as well as attacks; availability is security too.*

***Next video:** the goals that control who does what.*

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Video 2 · Controlling Who Does What

Security Fundamentals · Section 2

In this video, you will learn:

- Authentication, authorization, and accounting (AAA)
- What non-repudiation means
- How these support the CIA triad

AAA: who are you, what may you do, what did you do

*The CIA triad says what we protect. **AAA** says how we control who does what.*

Goal	Answers	Example
Authentication	"Who are you?"	Password, code, fingerprint, passkey
Authorization	"What may you do?"	A teller and a manager differ
Accounting	"What did you do?"	The logs of who did what, when

*Order: you **authenticate**, the system **authorizes**, then it **accounts** for what you did.*

Non-repudiation

Someone cannot credibly deny that they did something.

- *Sign a contract, and you cannot later claim you never did.*
- *Provided mainly by **digital signatures** plus solid **logs**.*
- *Matters most for **money, contracts, and legal records**.*

*It builds on **integrity** (the record was not changed) and **authentication** (we know who acted).*

Recap: controlling who does what

- ***Authentication:*** *prove who you are.* ***Authorization:*** *what you may do.* ***Accounting:*** *the record of what you did.*
- ***Non-repudiation:*** *you cannot deny you did it.*
- *These support the CIA triad; weak authentication breaks both confidentiality and integrity.*

Next video: *why the goals pull against each other, and how to balance them.*

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Video 3 · Balancing the Goals

Security Fundamentals · Section 2

In this video, you will learn:

- Why improving one goal can cost another
- How to choose a balance on purpose
- Using the goals as a checklist for any system

The goals pull against each other

*A good design is a **balance chosen on purpose**, not every goal maxed out.*

Push this up...	***...and this can suffer***
Confidentiality (more locks, more passwords)	*Availability and ease of use*
Availability (easy to reach)	*Confidentiality (more ways in)*
Integrity (approve every change)	*Speed and availability*
Security in general	*Usability and cost*

A bank that needed five passwords per transaction would be very "secure" and completely unusable.

The goals as a checklist

For any system, ask seven questions. Take an online shop:

Goal	Ask
Confidentiality	Who can read addresses and card details?
Integrity	Can anyone change a price or total?
Availability	What happens on the busiest day, or under attack?
Authentication	How do people prove who they are?
Authorization	Can a customer reach another's order?
Accounting	If money vanished, could we trace it?
Non-repudiation	Can a customer deny a purchase?

Recap: balancing the goals

- *The goals trade against each other and against usability; choose a balance based on **risk**.*
- *Spend protection where the asset is valuable and the threat is real.*
- *The seven goals are a checklist for thinking about any system.*

***Next lesson:** Threats, vulnerabilities, and risk, how we measure and manage the dangers these goals face.*